

AI Friday Season 2 – Problem Statements

Conversational AI Assistant for Retail Inventory Replenishment

PROBLEM STATEMENT

Retail store managers often struggle with timely inventory replenishment due to fragmented data sources and lack of real-time insights. Manual stock monitoring leads to overstocking or stockouts, negatively impacting sales and customer satisfaction. The volume of SKUs and frequent demand fluctuations complicate decision-making. Existing inventory management systems provide limited predictive capabilities and poor user experience, requiring technical expertise. Store managers need a simple, interactive assistant that understands natural language inquiries about inventory status and suggests replenishment actions. Current approaches fail to provide actionable recommendations or integrate external supplier lead times effectively. This gap causes lost revenue opportunities and increased carrying costs, emphasizing the need for an AI-powered conversational tool to streamline inventory decisions.

Data Considerations:

Requires access to SKU-level inventory data from internal retail databases in structured formats (CSV, SQL). Supplier lead times and catalog data accessible via REST APIs. Historical sales data for volume context. Data quality must be high to avoid incorrect recommendations. Privacy compliance involves limiting access to sensitive supplier contracts. Preprocessing includes demand trend summarization and stock threshold calculation. Synthetic data for development can simulate inventory levels and order cycles.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

The solution delivers an interactive chat-based UI where retail managers ask inventory questions and receive AI-generated, context-aware responses. Key features include real-time stock level querying, replenishment quantity recommendations, integration with supplier APIs for lead time checks, and alert notifications for low stock. The UI displays inventory dashboards with reorder suggestions and historical sales trends. KPIs include 90% query accuracy and 30% reduction in stockouts. The GenAI component interprets natural language queries and generates actionable insights. The prototype enables quick decision-making and order initiation directly from the interface.

Feel free to explore creative solutions that align with the problem context and objectives.

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AI-Enhanced Logistics Route Planner with Real-Time Traffic Integration

PROBLEM STATEMENT

Logistics companies face challenges optimizing delivery routes in real-time due to dynamic traffic conditions, vehicle constraints, and delivery time windows. Manual route planning is inefficient and often fails to adapt quickly to delays or incidents, leading to increased fuel costs, missed delivery deadlines, and reduced customer satisfaction. Existing route optimization tools lack AI-driven predictive capabilities and seamless integration with live traffic data. There is a need for an intelligent assistant that generates and updates optimized routes considering real-time factors and provides clear explanations to dispatchers, improving operational efficiency and delivery performance.

Data Considerations:

Real-time traffic and map data from APIs in JSON format; delivery order data including locations, time windows, and vehicle constraints from internal systems; historical delivery performance data for validation; synthetic delivery scenarios for testing; preprocessing to geocode addresses and normalize time formats; compliance with data privacy and third-party API usage terms; volume includes hundreds of daily delivery points.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

An interactive web application displaying AI-optimized delivery routes overlaid on maps with real-time traffic layers. Features include route adjustment controls, AI-generated textual summaries explaining route choices, and alerts for delays or route deviations. Dispatchers can export routes to vehicle navigation systems. Success metrics include delivery time reduction, route efficiency scores, and dispatcher user satisfaction. The prototype demonstrates multi-API integration, dynamic route generation, and user-friendly visualization tailored for logistics dispatch operations.

Feel free to explore creative solutions that align with the problem context and objectives.

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AI-Powered Travel Itinerary Optimizer for Logistics Coordinators

PROBLEM STATEMENT

Logistics coordinators in the travel industry face complex scheduling challenges when managing multi-stop trips involving various transportation modes and accommodations. Current manual planning methods are time-consuming and prone to errors, leading to costly delays and customer dissatisfaction. The volume of bookings and route variables has increased significantly, overwhelming existing tools that lack dynamic adaptability. Coordinators struggle with integrating real-time constraints such as transport availability, weather, and client preferences. Inefficient itinerary planning results in increased operational costs and suboptimal resource utilization. Existing software solutions often require heavy manual input and do not leverage AI to suggest optimized alternatives. This inefficiency impacts both customer experience and company profitability. There is a critical need for an intelligent assistant to generate optimized, feasible travel itineraries quickly and interactively to meet diverse client needs and logistical constraints.

Data Considerations:

Requires structured transport schedule data (flights, trains, buses) in JSON or API formats, accommodation availability data via hotel booking APIs, and client preference inputs in structured form. Real-time data integration from public transit and weather APIs is needed. Data volume is moderate but must be up-to-date and reliable. Preprocessing includes time zone normalization and conflict detection. Privacy compliance involves anonymizing client details in logs. Synthetic data generation for testing can include simulated bookings and route scenarios. Data quality checks are essential to avoid invalid itineraries.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

The application delivers an interactive web UI where logistics coordinators input trip parameters and receive optimized, multi-modal travel itineraries. Core features include AI-generated scheduling suggestions, editable itinerary timelines, real-time API integration for availability checks, and conflict alerts. The prototype demonstrates itinerary generation speed under 5 seconds and user editing with immediate re-optimization. KPIs include 85% user satisfaction and 50% reduction in planning time. The UI presents clear route maps, timelines, and exportable itinerary documents. Enhanced GenAI capabilities power itinerary reasoning and natural language explanations of scheduling choices.

Feel free to explore creative solutions that align with the problem context and objectives.

AI Friday Season 2 – Problem Statements

Generative AI Assistant for Retail Customer Support Chat

PROBLEM STATEMENT

Retail companies struggle with providing quick, accurate, and personalized customer service across multiple channels. Customers often experience long wait times and inconsistent answers due to limited agent availability and knowledge gaps. Current chatbot solutions lack deep understanding of product catalogs, promotions, and order statuses, leading to frequent escalations to human agents. This causes increased support costs, customer dissatisfaction, and lost sales opportunities. The volume of daily support queries can reach thousands, making manual handling impractical. There is a need for an intelligent assistant that understands retail context, can handle diverse queries, and seamlessly escalates complex issues. Improving first-contact resolution and response speed directly impacts customer retention and revenue.

Data Considerations:

Access to product catalog data in structured JSON or API format, including SKUs, descriptions, and promotions; order management system API to retrieve real-time order status and history; historic chat logs for intent understanding (synthetic data if real logs unavailable); user profile data respecting privacy policies; data preprocessing to anonymize PII and normalize product names; typical daily query volume simulated for load testing; integration with messaging platforms or custom web chat UI; compliance with GDPR for customer data.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

A web-based interactive chatbot interface powered by generative AI that answers FAQs, product queries, and order status questions in natural language. Features include dynamic fetching of product and order info via API calls, personalized upsell recommendations, and smooth escalation with conversation context passed to human agents. Dashboard displays chat session metrics, resolution rates, and user feedback. Success is measured by resolution accuracy, user satisfaction, and reduction in human agent load. Prototype showcases real-time API integration, conversational understanding, and a clean intuitive UI suitable for retail customers.

Feel free to explore creative solutions that align with the problem context and objectives.

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AI-Driven Supplier Risk Assessment Tool for Consumer Packaged Goods

PROBLEM STATEMENT

CPG companies depend on a network of suppliers whose operational risks such as financial instability, delivery delays, or compliance issues can disrupt supply chains. Currently, risk assessments are manual, periodic, and based on disparate data sources, limiting their effectiveness. There is a lack of AI-powered tools that synthesize supplier data, news, and market signals to provide up-to-date risk profiles and actionable insights. Improving supplier risk visibility helps companies proactively mitigate disruptions, maintain inventory levels, and avoid financial losses.

Data Considerations:

Supplier master data from ERP systems; external news feeds and financial data APIs (e.g., Bloomberg, Reuters); social media sentiment data; preprocessing for data normalization and entity resolution; synthetic risk event data for testing; compliance with data privacy and licensing; volume includes hundreds of suppliers monitored weekly.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

An interactive dashboard presenting supplier risk scores with AI-generated narrative explanations highlighting key risk factors. Users can drill down into supplier profiles, view historical risk trends, and export reports. Success metrics include risk assessment accuracy, user engagement, and impact on supply chain decisions. The prototype showcases multi-source data integration, GenAI summarization, and an intuitive UI designed for CPG supply chain managers.

Feel free to explore creative solutions that align with the problem context and objectives.

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AI-Powered Travel Itinerary Optimizer for Business Trips

PROBLEM STATEMENT

Frequent business travelers and travel managers face challenges in manually planning efficient multi-destination itineraries that balance cost, time, and preferences. Current planning often involves juggling multiple booking platforms and spreadsheets, leading to suboptimal routes, higher costs, and missed opportunities for consolidation. The coordination among flights, accommodations, and local transport lacks automation, causing delays and increased administrative workload. This impacts travel budgets and employee productivity, especially for companies managing hundreds of trips monthly. Existing travel tools rarely provide personalized, consolidated itinerary suggestions that dynamically adjust to constraints like meeting schedules or preferred airlines. There is a business need to reduce time spent on travel planning while ensuring traveler satisfaction and cost efficiency. Improving itinerary planning will reduce travel expenses, increase traveler convenience, and streamline approval workflows.

Data Considerations:

Flight schedules and pricing data from airline APIs (e.g., Amadeus, Skyscanner); hotel availability and pricing from accommodation APIs; user preferences and company travel policies stored in internal databases; calendar data for meeting schedules via Microsoft Graph or Google Calendar API; synthetic data generation for testing multi-leg trips and various constraints; data formats include JSON and CSV; privacy considerations for traveler personal info; basic preprocessing to normalize date-time formats and filter available options within budget constraints; ensure API rate limits are handled gracefully.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

An interactive web application where users input trip parameters and preferences to receive optimized multi-destination itineraries. Core features include AI-powered route suggestion leveraging generative language models to explain itinerary choices, integration with flight and hotel APIs for real-time availability, and an editable itinerary dashboard with drag-and-drop adjustment. Key deliverables are itinerary export to calendar and booking systems, cost and time savings metrics displayed in the UI, and feedback collection for continuous improvement. Success is measured by itinerary optimization improvements and user engagement metrics. The prototype demonstrates seamless API integration, real-time itinerary generation, and user-friendly interactive controls powered by GenAI explanations.

Feel free to explore creative solutions that align with the problem context and objectives.

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Real-Time Market Risk Alert System for Capital Markets

PROBLEM STATEMENT

Capital markets traders and risk managers need to monitor multiple risk factors continuously to mitigate losses due to market volatility. Currently, risk alerts are often delayed or buried in large data streams, leading to missed opportunities or late reactions. Manual monitoring is inefficient and error-prone, especially during high volatility periods. Existing automated systems lack contextual understanding and prioritization of alerts based on impact. There is a need for an AI-driven system that generates real-time, context-aware risk alerts with natural language explanations, enabling timely and informed decision-making.

Data Considerations:

Continuous streaming market price and volume data in JSON or CSV format from public or subscription APIs. Historical market risk event data for validation. Large data volume with low latency requirements. Preprocessing includes noise filtering and feature extraction. Compliance with data usage policies. Synthetic streaming data scenarios for stress testing.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

A web-based real-time risk alert dashboard powered by GenAI that analyzes market data streams and generates prioritized alerts with natural language explanations. Features include alert filtering, historical trend views, and API integration for live data. Success metrics: 90% alert relevance and 25% false positive reduction. Demo includes live data ingestion and alert generation. Documentation covers system design, API integration, and user instructions.

Feel free to explore creative solutions that align with the problem context and objectives.

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Automated Investment Portfolio Summary Generator for Capital Markets Clients

PROBLEM STATEMENT

Investment advisors spend significant time preparing portfolio summaries for clients, detailing holdings performance, asset allocation, and market commentary. These reports are often manually compiled from multiple systems, leading to delays and inconsistencies. Clients expect timely, clear, and personalized portfolio insights to make informed decisions. Current automation tools lack natural language generation capabilities tailored to financial data, resulting in static or generic reports. There is a need for a generative AI system that consolidates portfolio data and generates personalized, easy-to-understand summaries with actionable insights. This will improve advisor efficiency and client satisfaction.

Data Considerations:

Access to portfolio holdings data in structured formats (JSON, CSV), performance metrics, market data APIs, historical reports for language style reference, and client profiles. Data must comply with financial data privacy standards. Preprocessing involves data normalization and mapping to narrative elements. Synthetic portfolio data generation supports development.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

SOLUTION EXPECTATIONS

A web-based interactive application where advisors upload or connect to portfolio data sources and receive AI-generated personalized portfolio summaries. Features include language customization, annotations, and multi-format export. Integration with portfolio and market data APIs ensures up-to-date information. KPIs include 90% summary relevance, 60% reduction in report preparation time, and positive client feedback. The prototype delivers end-to-end report generation with editing and export capabilities.

Feel free to explore creative solutions that align with the problem context and objectives.